

Assignment-03

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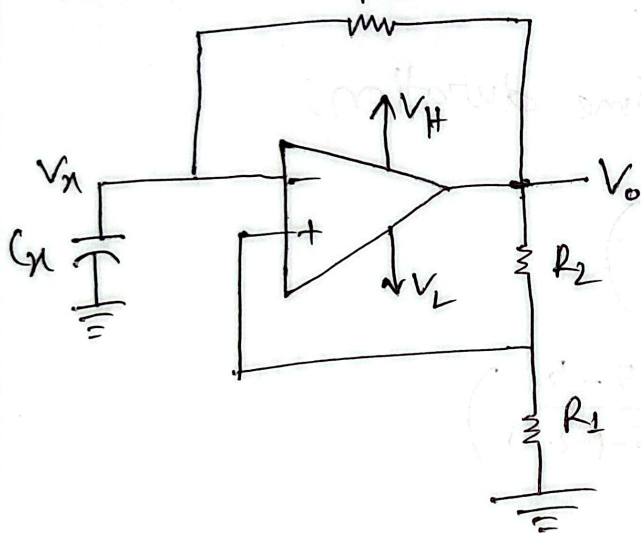
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* For the Schmitt trigger oscillator the saturation output voltages are $+10\text{ V}$ and -5 V . $R_1 = R_2 = 20\text{ k}\Omega$. $R_x = 50\text{ k}\Omega$ and $C_x = 0.01\text{ }\mu\text{F}$. Determine the frequency of oscillation and duty cycle.

Answer to the question

the circuit,



Given,

$$V_H = +10\text{ V}$$

$$V_L = -5\text{ V}$$

$$R_1 = 20\text{ k}\Omega$$

$$R_2 = 20\text{ k}\Omega$$

$$C_x = 0.01\text{ }\mu\text{F}$$

$$= 0.01 \times 10^{-6}\text{ F}$$

$$R_n = 50\text{ k}\Omega$$

Now,

$$V_{TH} = V_H \cdot \left(\frac{R_1}{R_1 + R_2} \right)$$

$$= 10 \cdot \left(\frac{20}{20 + 20} \right)$$

$$= 5\text{ V}$$

$$V_{TL} = V_L \cdot \left(\frac{R_1}{R_1 + R_2} \right)$$

$$= (-5) \cdot \left(\frac{20}{20 + 20} \right) = -2.5\text{ V}$$

and,

$$\tau = R_x \cdot C_x$$

$$= 50 \times 10^3 \times 0.01 \times 10^{-6}$$

$$= 0.5 \times 10^{-3} \text{ s}$$

So, T_1 is voltage high time duration,

$$T_1 = \tau \ln \left(\frac{V_H - V_{TL}}{V_H - V_{TH}} \right)$$

$$= 0.5 \times 10^{-3} \times \ln \left(\frac{10 - (-2.5)}{10 - 5} \right)$$

$$= 4.581 \times 10^{-4} \text{ s}$$

Similarly, T_2 is voltage low time duration,

$$T_2 = \tau \ln \left(\frac{V_L - V_{TH}}{V_L - V_{TL}} \right)$$

$$= 0.5 \times 10^{-3} \times \ln \left(\frac{-5 - 5}{-5 - (-2.5)} \right)$$

$$= 6.931 \times 10^{-4} \text{ s}$$

$$\therefore \text{Total period} = (4.581 \times 10^{-4} + 6.391 \times 10^{-4}) \text{ s}$$

$$\Rightarrow T = 1.151 \times 10^{-3} \text{ s}$$

$$\therefore \text{frequency} = \frac{1}{T} = \frac{1}{1.151 \times 10^{-3}}$$

$$\Rightarrow f = 868.589 \text{ Hz}$$

$$\text{Duty cycle} = \left(\frac{\text{High time}}{\text{High time} + \text{Low time}} \right) \times 100 \%$$

$$= \frac{T_1 \times 100}{T_1 + T_2} \%$$

$$= \frac{4.581 \times 10^{-4} \times 100}{1.151 \times 10^{-3}} \%$$

$$= 39.794 \%$$

Therefore, frequency of oscillation is 868.589 Hz and duty cycle is 39.794%.

(Ans.)